

Clinical Trial Protocol

Protocol Number: CTP-2026-0142

Study Title: A Phase II, Randomized, Double-Blind, Placebo-Controlled Study of Oral Compound AX-119 in Adult Patients with Moderate-to-Severe Rheumatoid Arthritis

Sponsor: Meridian Biotherapeutics Inc.

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1. Study Objectives

1.1 Primary Objective

The primary objective of this study is to evaluate the efficacy of AX-119 compared to placebo in reducing disease activity, as measured by the American College of Rheumatology 20% response criteria (ACR20), at Week 24 in adult patients with moderate-to-severe rheumatoid arthritis who have had an inadequate response to methotrexate.

1.2 Secondary Objectives

- To assess the safety and tolerability of AX-119 over 24 weeks of treatment.
- To evaluate the proportion of patients achieving ACR50 and ACR70 response at Week 24.
- To assess change from baseline in the Disease Activity Score (DAS28-CRP).
- To evaluate patient-reported outcomes using the Health Assessment Questionnaire (HAQ-DI).

2. Background and Rationale

Rheumatoid arthritis (RA) is a chronic, systemic autoimmune disease characterized by synovial inflammation, progressive joint destruction, and significant functional impairment. Despite the availability of conventional synthetic and biologic disease-modifying antirheumatic drugs (DMARDs), a substantial proportion of patients fail to achieve an adequate clinical response or experience loss of efficacy over time.

AX-119 is a novel, orally administered selective inhibitor of Janus kinase 1 (JAK1) designed to interrupt intracellular signaling pathways implicated in the inflammatory cascade of RA. In a completed Phase I study (CTP-2024-0087), AX-119 demonstrated a favorable pharmacokinetic profile and was well tolerated across single and multiple ascending dose cohorts, with no dose-limiting toxicities observed up to 200 mg once daily.

This Phase II study is designed to characterize the dose-response relationship of AX-119 and to generate preliminary efficacy and safety data to inform the design of subsequent Phase III trials.

3. Study Design

3.1 Overall Design

This is a multicenter, randomized, double-blind, placebo-controlled, parallel-group study. Approximately 240 patients will be randomized in a 1:1:1:1 ratio to receive AX-119 50 mg, AX-119 100 mg, AX-119 150 mg, or matching placebo, administered orally once daily for 24 weeks.

Randomization will be stratified by baseline methotrexate dose (≤ 15 mg/week vs. > 15 mg/week) and by geographic region. Patients who do not achieve at least a 20% improvement in tender and swollen joint counts at Week 12 will be eligible for early escape to open-label rescue therapy.

3.2 Study Duration

The total study duration for each patient is approximately 30 weeks, comprising a 2-week screening period, a 24-week double-blind treatment period, and a 4-week safety follow-up period after the last dose of study drug.

4. Study Population

4.1 Inclusion Criteria

- Male or female patients aged 18 to 75 years, inclusive.
- Diagnosis of rheumatoid arthritis per the 2010 ACR/EULAR classification criteria, with disease duration of at least 6 months.
- Moderate-to-severe disease activity at screening and baseline, defined as a DAS28-CRP score greater than 3.2.
- Inadequate response to methotrexate therapy for at least 12 weeks prior to screening, at a stable dose for at least 4 weeks.
- Willingness and ability to comply with scheduled visits, treatment plans, laboratory tests, and other study procedures.

4.2 Exclusion Criteria

- Prior exposure to any JAK inhibitor, including but not limited to tofacitinib, baricitinib, or upadacitinib.
- Active or latent tuberculosis without documented completion of appropriate treatment.
- History of malignancy within the past 5 years, except adequately treated non-melanoma skin cancer or cervical carcinoma in situ.
- Estimated glomerular filtration rate (eGFR) less than 40 mL/min/1.73m² at screening.
- Pregnant or breastfeeding, or planning to become pregnant during the study period.
- Concurrent participation in another interventional clinical trial within 30 days prior to screening.

5. Treatment Plan

5.1 Study Drug Administration

Study drug will be supplied as identical-appearing film-coated tablets and administered orally once daily, in the morning, with or without food. Patients will be instructed to take the study drug at approximately the same time each day and to record dosing in an electronic diary.

5.2 Concomitant Medications

Patients must continue background methotrexate at a stable dose throughout the study. Stable doses of oral corticosteroids (equivalent to prednisone 10 mg/day or less) and non-steroidal anti-inflammatory drugs (NSAIDs) are permitted. Live vaccines are prohibited from screening through the end of the safety follow-up period.

6. Efficacy Assessments

The primary efficacy endpoint, ACR20 response at Week 24, will be assessed by a blinded joint assessor at each scheduled visit. Tender and swollen joint counts (68/66 joints), patient and physician global assessments, patient pain assessment, C-reactive protein (CRP), and the Health Assessment Questionnaire Disability Index (HAQ-DI) will be collected at Screening, Baseline, and Weeks 2, 4, 8, 12, 16, 20, and 24.

Radiographic assessment of hands and feet will be performed at Baseline and Week 24 to evaluate structural joint damage using the modified Total Sharp Score (mTSS), read centrally by independent radiologists blinded to treatment assignment and visit order.

7. Safety Assessments

Safety will be monitored throughout the study through physical examinations, vital signs, clinical laboratory tests (hematology, chemistry, lipid panel, urinalysis), and 12-lead electrocardiograms. Adverse events will be recorded from the time of informed consent through the end of the safety follow-up period and graded according to the National Cancer Institute Common Terminology Criteria for Adverse Events (NCI-CTCAE), version 5.0.

Special attention will be given to adverse events of interest associated with the JAK inhibitor class, including serious infections, herpes zoster reactivation, venous thromboembolism, malignancy, and major adverse cardiovascular events (MACE). An independent Data Safety Monitoring Board (DSMB) will review unblinded safety data at prespecified intervals.

8. Statistical Considerations

8.1 Sample Size

A total sample size of 240 patients (60 per arm) provides approximately 90% power to detect a 20-percentage-point difference in ACR20 response rate between the highest AX-119 dose group and placebo, assuming a placebo response rate of 30% and a two-sided alpha of 0.05, accounting for an estimated 10% dropout rate.

8.2 Statistical Analysis

The primary analysis will compare ACR20 response rates at Week 24 between each AX-119 dose group and placebo using a Cochran-Mantel-Haenszel test stratified by randomization factors. Patients who discontinue treatment or receive rescue therapy prior to Week 24 will be classified as non-responders under the non-responder imputation approach. A hierarchical testing procedure will be used to control the family-wise Type I error rate across dose comparisons.

9. Ethical Considerations

This study will be conducted in accordance with the ethical principles originating from the Declaration of Helsinki and consistent with Good Clinical Practice (GCP) and applicable regulatory requirements. The protocol, informed consent form, and any amendments will be reviewed and approved by an Institutional Review Board (IRB) or Independent Ethics Committee (IEC) prior to enrollment of any patient at a given site. Written informed consent will be obtained from each patient prior to the initiation of any study-related procedures.

10. Data Management

All study data will be collected using a validated electronic data capture (EDC) system compliant with 21 CFR Part 11. Source documents at each site will be maintained and made available for monitoring, audit, and inspection. Data queries will be resolved by site personnel within 5 business days of issuance. Database lock will occur only after all queries are resolved and the data have been reviewed and approved by the sponsor's biostatistics and clinical data management teams.

Randomization codes will be maintained by an independent, unblinded statistician and will not be accessible to the study team, investigators, or sponsor personnel involved in study conduct until database lock, except in cases of medical emergency requiring emergency unblinding, which will follow the procedure described in the site pharmacy manual.

11. Study Committees and Oversight

11.1 Data Safety Monitoring Board

An independent Data Safety Monitoring Board (DSMB), composed of clinicians and a biostatistician with no other involvement in the study, will meet at prespecified intervals to review unblinded safety data and make recommendations regarding study continuation, modification, or termination based on emerging safety signals.

11.2 Steering Committee

A Steering Committee comprising the sponsor's medical monitor and external rheumatology experts will provide scientific oversight of the trial, review protocol amendments, and advise on the interpretation of interim data summaries that do not include treatment assignment.

12. Withdrawal and Discontinuation Criteria

Patients may withdraw from the study at any time, for any reason, without prejudice to future care. Investigators may discontinue a patient's participation for safety reasons, lack of compliance, use of a prohibited concomitant medication, or investigator judgment that continued participation is not in the patient's best interest. All patients who discontinue study drug prematurely will be encouraged to complete the safety follow-up visit and end-of-study assessments whenever feasible, since these data are important for the overall safety evaluation of AX-119.

13. Appendix: Schedule of Assessments (Summary)

Screening assessments occur within 14 days prior to randomization and include demographics, medical history, physical examination, vital signs, 12-lead ECG, clinical laboratory tests, tuberculosis screening, and pregnancy testing where applicable. Baseline assessments, conducted on Day 1 prior to first dose, include joint counts, patient and physician global assessments, HAQ-DI, and randomization. Efficacy and safety assessments are then repeated at Weeks 2, 4, 8, 12, 16, 20, and 24, with radiographic imaging repeated only at Week 24. A safety follow-up visit occurs 4 weeks after the last dose of study drug or early discontinuation.